Prompt harmfulness classification

How do representation and classifier architecture affect macro-F1?

Training population	Rows
Original WildGuardTrain	86,759
After category exclusion	83,739
After cleanup and deduplication	46,692
Balanced CV sample	3,000

Exclude `toxic_language_hate_speech` first. Cleanup removes 14 empty prompts, 37,003 duplicate rows and 30 rows with conflicting labels.

Input: prompt text only
Target: unharmful (0) or harmful (1)

Evaluation: five shared stratified folds.
Nine systems, 45 fits, seed 42.
1,500 prompts per class.

Primary: macro-F1.
Secondary: harmful recall, benign false-positive rate and accuracy.

The sample is 50% harmful, compared with 51.58% before deduplication. **The official test remains untouched.**

Sources: data_audit.json, settings.json. Results use only filtered WildGuardTrain. 3 November 2026.

Representations and classifier architectures

A fresh BPE tokenizer and IDF fit only the training portion of each fold

Representation	Information retained
Indicators	Presence or absence of each token
Token counts	Number of occurrences
TF-IDF	Counts weighted by training rarity

Classifier	Structure
Logistic	Linear logits
One-layer MLP	64 ReLU units
Two-layer MLP	64 then 32 ReLU units

Shared features: Metaspace BPE, 1,500 tokens, minimum frequency 2. TF-IDF uses smoothed IDF and L2 normalization. All representations lose token order.

Comparable neural training: 10 fixed epochs, dropout 0.2, Adam learning rate 0.001, batch size 64. Validation folds do not select checkpoints.

Source: project1.ipynb and utils/p1/functions.py. Reuse: course BPE, SparseMLP, training helpers and Bayesian model.

Token counts with one hidden layer lead the means

Macro-F1 averaged over five folds, shown with fold standard deviation

Classifier	Indicators	Token counts	TF-IDF
Logistic regression	0.7398 +/- 0.0207	0.7429 +/- 0.0247	0.7665 +/- 0.0132
One-layer MLP	0.7666 +/- 0.0195	**0.7786** **+/- 0.0156**	0.7708 +/- 0.0132
Two-layer MLP	0.7697 +/- 0.0165	0.7762 +/- 0.0141	0.7558 +/- 0.0202

Highest observed mean: one-layer MLP with counts, 0.7786. The two-layer count model is close at 0.7762.

Extra depth shows no consistent gain. The highest mean does not establish a decisive winner.

Source: p1_summary.csv. Fold SD is descriptive, not an independent-fold confidence interval.

Bayesian evidence for the preselected comparisons

Mean difference B - A. ROPE +/-0.01 macro-F1 defines practical equivalence.

A: Logistic indicators

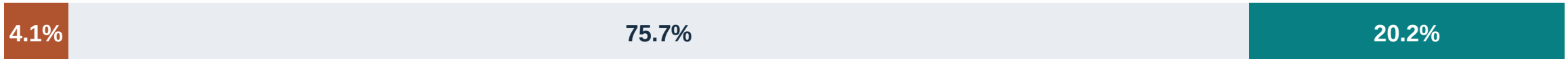
B: Logistic TF-IDF

Mean B - A: +0.0268

A: Logistic TF-IDF

B: One-layer MLP TF-IDF

Mean B - A: +0.0042



A better beyond ROPE Practically equivalent B better beyond ROPE

TF-IDF has moderate evidence of improving logistic regression. On TF-IDF, the evidence mostly favours practical equivalence between logistic regression and the one-layer MLP.

Source: p1_rope.csv. Tutorial correlated-t posterior, one five-fold run. Probabilities depend on its assumptions.

The best choice depends on the error tradeoff

Counts plus a shallow MLP is the strongest observed F1 baseline. Logistic TF-IDF is a useful simpler alternative.

System	Macro-F1	Harmful recall	Benign FPR
One-layer MLP + counts	0.7786	74.9%	19.1%
Logistic + TF-IDF	0.7665	77.7%	24.4%

Counts MLP: fewer benign false alarms.
Logistic TF-IDF: catches more harmful prompts.

High confidence can be wrong.
Counts MLP: 286 false positives and 377 false negatives. 366 of its 663 errors had confidence >=0.8.

Conclusion: representation and architecture should be chosen together. Extra depth is not justified by a consistent gain here, and confidence alone is insufficient for reliable decisions.

Limits: one balanced sample, five overlapping folds, label noise and related prompts. Official test untouched. No deployed-safety claim.

Sources: p1_summary.csv, p1_oof_predictions.csv. Code adapted from the course tutorial with disclosed AI assistance.